

WhatsApp CRM & Messaging Engine

Standalone SaaS + Optional CRM-Integrated Module

Version 1.0

Status: Draft

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the WhatsApp CRM & Messaging Engine.

The module shall operate in two modes:

1. Standalone WhatsApp Engagement Platform
2. Integrated Module within the Planned SaaS CRM Suite

The system enables WhatsApp as a structured, compliant, automation-ready conversational channel for lead generation, qualification, campaign execution, support, and revenue acceleration.

This is not a messaging tool.

It is a conversion engine with auditability, AI governance, and role-based control.

1.2 Product Positioning

The module supports:

- Lead Capture via WhatsApp
- AI + Human Hybrid Conversations
- Campaign Orchestration
- Workflow Automation
- Compliance Enforcement
- Performance Reporting

When integrated with CRM:

- Leads sync to CRM
 - Conversations attach to CRM timeline
 - Pipeline stages update automatically
 - Lead scoring integrates
 - Deal tracking supported
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2. Scope

2.1 In Scope

- Inbound WhatsApp Messaging
 - Outbound Messaging (session + template)
 - Lead Auto-Creation
 - Campaign Broadcasting
 - AI-driven Conversations
 - Human Escalation
 - Automation Triggers
 - Role-Based Access
 - Consent Tracking
 - Reporting
 - Public API & Webhooks
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3. System Modes

3.1 Standalone Mode

System maintains:

- Internal Lead Database
- Internal Timeline
- Internal Pipeline
- Internal User Roles
- Internal Campaign Engine

3.2 CRM-Integrated Mode

If CRM subscription active:

- Leads bind to CRM module
 - Timeline writes to CRM Lead Activity Log
 - Stage changes sync
 - AI context pulls CRM data
 - Campaign outcomes affect CRM scoring
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4. WhatsApp Account & Tenant Model

- Tenant-scoped identity through meta
- BYO credentials/ apply through platform

Meta verification outside the system

Our system will give access to submit information and rest meta will take care

Each conversation must bind to:

- Tenant ID
 - Lead ID
 - Assigned Owner
 - Campaign ID (optional)
 - Conversation Ownership Type (AI/Human/Hybrid)
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5. Lead Capture & Binding

5.1 Inbound Lead Creation

When a new WhatsApp message is received:

System shall:

- Match by phone number

- If no match → auto-create lead
- Capture metadata
- Tag source = WhatsApp
- Record first-touch timestamp

User Stories

As an Agent

I want incoming WhatsApp messages to auto-create leads
So that no inquiry is missed.

As a Manager

I want lead source tagging automated
So that channel attribution is accurate.

5.2 Lead Enrichment

System shall allow:

- AI qualification questions
- Field updates via conversation
- Tag application
- Stage updates
- Pipeline movement

In CRM mode:

Changes must reflect in CRM Lead Object.

User Stories

As an Agent

I want to update lead stage directly from chat
So that workflow remains centralized.

As a Manager

I want AI to auto-qualify using structured logic
So that high-intent leads are prioritized.

6. Messaging Capabilities

6.1 Supported Message Types

- Text
- Image
- Video
- Audio
- Documents
- Location
- Interactive Buttons
- List Messages
- CTA Buttons
- Template Messages

6.2 Messaging Rules

- **Free-text allowed only within 24-hour session window :**
<https://business.whatsapp.com/products/platform-pricing>
- Template required to initiate conversation
- Template approval lifecycle enforced
- Rate limits applied per tenant

User Stories

As an Agent

I want to send interactive CTAs

So that customers can take direct action.

As an Admin

I want template approval workflow enforced

So that compliance risk is minimized.

7. AI Conversation Engine

7.1 AI Capabilities

AI may:

- Reply to inbound messages
- Ask structured qualification questions

- Handle FAQs
- Schedule callbacks
- Trigger campaigns
- Escalate to human
- Update lead fields

AI may NOT:

- Send unapproved templates
- Modify billing data
- Override compliance restrictions
- Loop infinite responses

7.2 AI Context Injection

AI context sources:

- Lead data
- Past WhatsApp messages
- Call summaries
- Campaign objectives
- Tenant compliance rules

All AI context injection must be logged and auditable.

User Stories

As a Manager

I want AI responses governed by compliance rules
So that brand risk is controlled.

As an Agent

I want AI suggestions visible but editable
So that control remains human-led.

8. Human + AI Handoff Model

8.1 Ownership Modes

- AI-Owned

- Human-Owned
- Hybrid Assist

Switching ownership must:

- Be explicit
- Be logged
- Reset SLA timers

8.2 Escalation Triggers

- Customer request
- AI low-confidence score
- Negative sentiment detection
- Compliance-sensitive query

User Stories

As an Agent

I want full transcript visible when AI escalates
So that context is preserved.

As a Supervisor

I want ability to freeze AI replies
So that escalation can be controlled.

9. Campaign Engine

9.1 Campaign Types

- Broadcast → verified broadcast list
- Drip Sequence
- Event-Based Trigger
- AI Follow-up Campaign
- Call-to-WhatsApp

9.2 Campaign Rules

- Template-only initiation

- Opt-in enforcement
- Daily send caps
- Time window restrictions
- Budget guardrails

Campaign outcomes must update:

Standalone Mode:

- Lead Stage
- Lead Score
- Timeline Entry

CRM Mode:

- CRM Lead Stage
- CRM Score
- CRM Deal Object (if exists)

User Stories

As a Manager

I want broadcast campaigns limited by opt-in rules
So that compliance is enforced automatically.

As a Revenue Lead

I want drip campaigns tied to lead stage
So that nurturing aligns with pipeline position.

10. Template Management

Lifecycle:

- Draft
- Submitted
- Approved
- Rejected
- Archived

Template attributes:

- Language

- Category
- Variables
- CTA buttons
- Versioning

All template usage logged per:

- Lead
 - Campaign
 - Tenant
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11. Automation & Workflow Triggers

WhatsApp events may trigger:

- Lead Stage Change
- Task Creation
- Callback Scheduling
- AI Call Initiation
- Campaign Enrollment
- SLA Escalation

Triggers must be:

- Configurable
- Role-restricted
- Auditable

User Stories

As an Operations Head
I want WhatsApp replies to trigger pipeline progression
So that system operates event-driven.

12. Activity Timeline

Every WhatsApp event must appear in timeline:

- Sent messages
- Received messages
- AI responses
- Human responses
- Media shared
- Campaign touches
- Stage changes

Timeline is:

- Chronological
- Immutable
- Read-only

CRM Mode:

Timeline syncs to CRM Activity Log.

13. Search & Filters

Filters:

- Lead Stage
- Owner
- Unread Status
- SLA Breach
- Campaign
- Tags
- AI vs Human handled

Saved Views supported per user role.

14. Permissions & Security

14.1 Role-Based Permissions

Permissions include:

- View Conversations

- Reply Manually
- Approve Templates
- Enable/Disable AI
- Export Conversations
- Create Campaigns
- Configure Automations

All enforced via RBAC engine.

14.2 Compliance & Privacy

System must:

- Track consent timestamp
 - Enforce opt-out immediately
 - Block messaging to non-consented users
 - Respect retention policy per tenant
 - Support data deletion request
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15. Cost Control & Safeguards

System must include:

- Rate limiting
- Daily caps per campaign
- AI throttling
- Infinite loop detection
- Forced AI pause logic
- Tenant-level spending guardrails

No uncontrolled WhatsApp spend allowed.

16. Reporting & Analytics

System emits:

- Message Delivery Status
- Read Rates

- Response Rates
- Conversion Events
- AI vs Human Performance
- Campaign Effectiveness
- Lead Velocity Metrics
- Revenue Attribution (CRM Mode)

Dashboards available in Reporting Module.

17. API & Third-Party Integration

System shall expose:

- **REST API**
- **Webhooks**
- Auth via OAuth/API Keys
- Versioned endpoints

Supported external integration:

- Third-party CRM sync
- ERP systems
- Marketing automation tools
- Custom analytics

Events exposed:

- New Lead
 - New Message
 - Campaign Sent
 - Stage Updated
 - Opt-In Changed
 - SLA Breach
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18. Non-Functional Requirements

- 99.9% uptime
- Horizontal scalability

- Secure multi-tenant isolation
 - Encryption in transit & at rest
 - Audit logs retained minimum 12 months
 - Sub-second UI response time
 - AI guardrails enforced at runtime
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19. Future Enhancements

- Multi-language AI routing
 - WhatsApp Commerce Catalog integration
 - Revenue prediction modeling
 - AI intent clustering
 - Cross-channel orchestration engine
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1. Whatsapp workflow data collection → campaign → templates based and promotional
2. Centralised system : payment, reports, → transactional
3. Communication on CRM at one place
4. Whatsapp catalog
5. Whatsapp communication
6. Whatsapp AI agent
7. Whatsapp business is must

Meta Business api

Business number

Subscription : info

Templates bante ho

Approval from meta

App wo templates send kr skte ho in a campaign

Agar user ne 24 hours me reply kiya then you can send message

Template followup

- direc

